AWS Cloud Implementation - Big Data, GenAI & Agentic AI Architecture

Data sources

Data ingestion

Batch data
AWS DataSync
Transfer Family · AWS Glue

Real-time streaming
Amazon Kinesis Data Streams
Amazon MSK (Kafka)

Change data capture
AWS DMS · Zero-ETL
Aurora/RDS to Redshift

Reverse ETL
Amazon AppFlow
AWS Glue · EventBridge

Data Ocean

Data lake
Amazon S3 + Lake Formation

Data pond
S3 domain bucket + LF tags

Data puddle
S3 project bucket + IAM

Data mesh
Amazon DataZone

Data contracts
Glue Schema Registry

Workflow manager · Amazon MWAA (Airflow) · AWS Step Functions · EventBridge Scheduler

Data transformation

Data labeling
SageMaker Ground Truth

Query engines
Athena · EMR · Redshift Spectrum

Data sci. libraries
SageMaker Studio · Glue Notebooks

Data enrichment
AWS Data Exchange · Glue

AI infrastructure

Knowledge store

Feature store
SageMaker Feature Store

Vector database
OpenSearch · Aurora pgvector

Knowledge graph
Amazon Neptune

LLMOps + MLOps

Model registry
SageMaker Model Registry

Prompt registry
Bedrock Prompt Mgmt

Fine-tuning · RLHF
Bedrock Customization · SageMaker

Eval & red-teaming
Bedrock Evaluations · Clarify

CI/CD · experiment tracking
CodePipeline · CodeBuild · MLflow

AI inference

Online predictor
SageMaker Real-time Endpoints

Batch predictor
SageMaker Batch Transform · AWS Batch

Streaming inference
Lambda + Kinesis · Managed Flink

Edge inference
AWS IoT Greengrass · SageMaker Edge

AI gateway
Bedrock + API Gateway · CloudFront

GenAI platform

Guardrails & orchestration
Bedrock Guardrails · Bedrock Agents

Model & prompt hub
Bedrock · SageMaker JumpStart

Custom knowledge model
Bedrock Custom Models · JumpStart

Foundation models
Amazon Bedrock
Claude · Llama · Titan · Nova · Mistral

RAG pipeline
Bedrock Knowledge Bases
OpenSearch · Aurora pgvector

Multimodal
Rekognition · Transcribe
Textract · Polly · Bedrock Nova

Agentic AI layer

Agent orchestrator
Bedrock Agents
AgentCore Runtime
Step Functions

Tool registry / MCP
AgentCore Gateway
AWS Lambda · API Gateway

Agent memory
AgentCore Memory
DynamoDB · ElastiCache
OpenSearch

HITL interface
Amazon Augmented AI (A2I)
SageMaker Ground Truth

Agent monitoring
AgentCore Observability
CloudWatch · X-Ray

Analytics & BI

Data viz & storyboard
Amazon QuickSight
Managed Grafana

Data search
Amazon Kendra · OpenSearch

Data accumulation
Redshift · Aurora · ElastiCache

Content analysis
Comprehend · Textract
Translate · Transcribe

Decision management
Step Functions · SageMaker
Personalize · Forecast

Agentic analytics
Amazon Q in QuickSight
Q Business · Q Developer

AI trust, governance & security

Data governance
Amazon DataZone · Lake Formation · Glue Data Catalog
Macie (PII) · KMS · CloudTrail · IAM Identity Center · AWS Config
Lineage via DataZone · Entity Resolution · GuardDuty

AI governance
Bedrock Guardrails · SageMaker Model Cards · Clarify
Model Monitor · AWS Audit Manager · CloudTrail
Responsible AI dashboards · Bedrock Logging

Assessment framework

AI maturity assessment
Well-Architected (ML Lens) · Trusted Advisor

Economic benefit assessment
Cost Explorer · Pricing Calculator · Budgets

Responsible AI KPIs
SageMaker Model Monitor · Clarify
CloudWatch Metrics · Bedrock Evaluations

Applications

Data layers | AI/ML infrastructure | GenAI & analytics | Agentic AI | Governance & trust | Assessment